of a change in nature's laws than do the differences you find when performing your gymnastics routine on earth or the moon.[2]

These examples give some sense of why many consider, and I suspect Feynman would have agreed, that the copious symmetries underlying natural law present a close runner-up to the atomic hypothesis as a summary of our deepest scientific insights. But there is more to the story. Over the last few decades, physicists have elevated symmetry principles to the highest rung on the explanatory ladder. When you encounter a proposed law of nature, a natural question to ask is: Why this law? Why special relativity? Why general relativity? Why Maxwell's theory of electromagnetism? Why the Yang-Mills theories of the strong and weak nuclear forces (which we'll look at shortly)? One important answer is that these theories make predictions that have been repeatedly confirmed by precision experiments. This is essential to the confidence physicists have in the theories, certainly, but it leaves out something important.

Physicists also believe these theories are on the right track because, in some hard-to-describe way, they *feel* right, and ideas of symmetry are essential to this feeling. It feels right that no location in the universe is somehow special compared with any other, so physicists have confidence that translational symmetry should be among the symmetries of nature's laws. It feels right that no particular constant-velocity motion is somehow special compared with any other, so physicists have confidence that special relativity, by fully embracing symmetry among all constant-velocity observers, is an essential part of nature's laws. It feels right, moreover, that *any* observational vantage point—regardless of the possibly accelerated motion involved—should be as valid as any other, and so physicists believe that general relativity, the simplest theory incorporating this symmetry, is among the deep truths governing natural phenomena. And, as we shall shortly see, the theories of the three forces other than gravity— electromagnetism and the strong and weak nuclear forces—are founded on other, somewhat more abstract but equally compelling principles of symmetry. So the symmetries of nature are not merely consequences of nature's laws. From our modern perspective, symmetries are the foundation from which laws spring.

Symmetry and Time